

Subject: Fundamentals of Engineering Mathematics -II**Module 1 : Differential Equations of First Order and First Degree****Bernoulli's Equation**

The Bernoulli's Differential Equation is a first order differential equation which can be converted to the linear form by a simple substitution. Once the Bernoulli's Differential Equation is converted to the linear form, its solution is obtained using the new variables and finally the obtained answer is changed back to the original given variables.

Form 1: $\frac{dy}{dx} + P(x)y = Q(x)y^n$ and

Form 2: $\frac{dx}{dy} + P'(y)x = Q'(y)x^n$

where P and Q are functions of x alone or constant;
and P' and Q' are functions of y alone or constant.

Examples:

1. $\frac{dy}{dx} + y = x^3 y^3$
2. $\frac{dx}{dy} + \cos y \cdot x = x^{3/2}$

Method for solving Bernoulli's Equation: Given Bernoulli's equation is,

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

divide by ' y^n ' we get,

$$\frac{1}{y^n} \frac{dy}{dx} + \frac{P(x)}{y^{n-1}} = Q(x) \text{---(1)}$$

put $\frac{1}{y^{n-1}} = w$

Differentiating both sides w.r.t x we get,

$$\frac{d}{dx}y^{1-n} = \frac{dw}{dx} \Rightarrow (1-n)y^{-n}\frac{dy}{dx} = \frac{dw}{dx} \Rightarrow \frac{1}{y^n}\frac{dy}{dx} = \frac{1}{(1-n)}\frac{dw}{dx}$$

Hence equation (1) becomes,

$$\frac{1}{(1-n)}\frac{dw}{dx} + P(x)w = Q(x)$$

Therefore,

$$\frac{dw}{dx} + (1-n)P(x)w = Q(x)(1-n)$$

Which is a linear D.E. and hence can be solved and finally in the solution replace 'w' by $\frac{1}{y^{n-1}}$

Examples:

1. Solve $x\frac{dy}{dx} + y = x^3y^6$

Solution: Given DE is $x\frac{dy}{dx} + y = x^3y^6$;

Divide by xy^6 we get

$$\frac{1}{y^6}\frac{dy}{dx} + \frac{1}{x}\frac{1}{y^5} = x^2 \text{---(1)}$$

put $\frac{1}{y^5} = w$,

differentiating this w.r.t. x we get,

$$-5y^{-5-1}\frac{dy}{dx} = \frac{dw}{dx} \Rightarrow \frac{-5}{y^6}\frac{dy}{dx} = \frac{dw}{dx} \Rightarrow \frac{1}{y^6}\frac{dy}{dx} = -\frac{1}{5}\frac{dw}{dx}$$

so, eq(1) becomes,

$$-\frac{1}{5}\frac{dw}{dx} + \frac{1}{x}w = x^2$$

$$\therefore \frac{dw}{dx} - \frac{5}{x}w = -5x^2 \text{---(2)}$$

eq(2) is linear DE with $P(x) = \frac{-5}{x}$ and $Q(x) = -5x^2$

Therefore

$$\begin{aligned} I.F. &= e^{\int P(x)dx} = e^{\int \frac{-5}{x} dx} \\ &= e^{-5 \log x} = e^{\log x^{-5}} = x^{-5} = \frac{1}{x^5} \end{aligned}$$

Therefore the G.S. of eq(2) is

$$\begin{aligned} w \cdot (I.F.) &= \int Q(x) \cdot (I.F.)dx + c \\ w \frac{1}{x^5} &= \int (-5x^2) \left(\frac{1}{x^5} \right) dx + c \\ \frac{w}{x^5} &= -5 \int x^{-3} dx + c \\ &= (-5) \left(\frac{x^{-3+1}}{-3+1} \right) + c \\ &= \frac{5}{2x^2} + c \\ \therefore w &= \frac{5}{2}x^3 + cx^5 \\ \text{therefore. } \frac{1}{y^5} &= \frac{5}{2}x^3 + cx^5 \end{aligned}$$

2. Solve $xy(1 + xy^2)\frac{dy}{dx} = 1$

Solution: Given DE is $xy(1 + xy^2)\frac{dy}{dx} = 1$

$$\begin{aligned} \frac{dx}{dy} &= xy + x^2y^3 \\ \frac{dx}{dy} - xy &= x^2y^3 \\ \therefore \frac{1}{x^2} \frac{dx}{dy} - \frac{1}{x}y &= y^3 \text{---(1)} \end{aligned}$$

put $-\frac{1}{x} = w$,
 differentiating this w.r.t. y we get,

$$-\frac{1}{x^2} \frac{dx}{dy} = \frac{dw}{dy} \Rightarrow \frac{1}{x^2} \frac{dx}{dy} = \frac{dw}{dy}$$

so, eq(1) becomes,

$$\frac{dw}{dy} + yw = y^3 \quad \text{---(2)}$$

eq(2) is linear DE with $P(y) = y$ and $Q(y) = y^3$

Therefore $I.F. = e^{\int P(y)dy} = e^{\int y dy} = e^{\frac{y^2}{2}}$

Therefore the G.S. of eq(2) is

$$\begin{aligned} w \cdot (I.F.) &= \int Q(y) \cdot (I.F.) dy + c \\ we^{\frac{y^2}{2}} &= \int (y^3)(e^{\frac{y^2}{2}}) dy + c \end{aligned}$$

$$\text{put } \frac{y^2}{2} = t \Rightarrow \frac{2y}{2} dy = dt \Rightarrow y dy = dt$$

$$\begin{aligned} we^t &= \int (2t)(e^t) dt + c \\ &= 2[te^t - e^t] + c \\ \frac{-e^{\frac{y^2}{2}}}{x} &= 2\left[\frac{y^2}{2}e^{\frac{y^2}{2}} - e^{\frac{y^2}{2}}\right] + c \end{aligned}$$

Therefore the General solution is given by

$$\frac{-e^{\frac{y^2}{2}}}{x} = 2\left[\frac{y^2}{2}e^{\frac{y^2}{2}} - e^{\frac{y^2}{2}}\right] + c$$

Practice Problems:

1. $\sin 2x \frac{dy}{dx} = y + \tan x$

2. $(y + 1)dx + (x - (y + 2)e^y)dy = 0$

3. $dr + (2r \cos \theta + \sin 2\theta)d\theta = 0$

4. $\frac{dy}{dx} + \frac{4x}{x^2 + 1}y = \frac{1}{(x^2 + 1)^3}$

5. $y \frac{dx}{dy} = x - yx^2 \sin y$

6. solve $\frac{dy}{dx} + \frac{x}{1-x^2}y = x\sqrt{y}$

7. Solve: $\frac{dy}{dx} = 1 - x(y - x) - x^3(y - x)^2$

8. Solve: $4x^2y \frac{dy}{dx} = 3x(3y^2 + 2) + (3y^2 + 2)^3$